Chapter 11. Cluster Analysis

K-means, K-medoids, hierarchical clustering, Gaussian mixtures and the EM algorithm, choosing K

Contents

1 Dissimilarity and Within-Cluster Scatter **2**

1.1 Limits of combinatorial algorithms . . . 2

2 K-means and K-medoids **2**

2.1 K-means . . . 3

2.2 K-medoids . . . 3

3 Hierarchical Clustering **4**

3.1 Agglomerative . . . 4

3.2 Divisive . . . 5

4 Gaussian Mixtures and the EM Algorithm **5**

4.1 Model . . . 5

4.2 The EM algorithm ($K = 2$) . . . 5

5 Choosing K and Other Topics **6**

6 Selected exercises **7**

Unlike classification (Ch. 10), clustering groups similar observations without labels (unsupervised learning). A clustering is an encoder $C : \{1, \dots, n\} \to \{1, \dots, K\}$.

1 Dissimilarity and Within-Cluster Scatter

Motivation. Clustering looks for groups without labels, so we must decide what makes a grouping good. Small within-cluster scatter is the usual criterion; because total scatter is fixed, this is the same as large between-cluster scatter.

Definition 11.1: within-cluster scatter

For a dissimilarity $d(i, j) \geq 0$,

$$W(C) = \frac{1}{2} \sum_{k=1}^{K} \frac{1}{n_k} \sum_{i:C(i)=k} \sum_{j:C(j)=k} d(i, j), \qquad n_k = \#\{i : C(i) = k\}.$$

Smaller W is better.

Theorem 11.2: with squared Euclidean distance $W(C)$ is the within-cluster sum of squares

If $d(i, j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, then

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2, \qquad \bar{\mathbf{x}}_{(k)} = \frac{1}{n_k} \sum_{C(i)=k} \mathbf{x}_i,$$

and $T = \sum_i \|\mathbf{x}_i - \bar{\mathbf{x}}\|^2 = W(C) + B(C)$ with $B(C) = \sum_k n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$.

Proof [Supplement]. Within cluster k, $\mathbf{x}_i - \mathbf{x}_j = (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) - (\mathbf{x}_j - \bar{\mathbf{x}}_{(k)})$ and $\sum_{C(i)=k} (\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}) = \mathbf{0}$, so

$$\sum_{C(i)=k} \sum_{C(j)=k} \|\mathbf{x}_i - \mathbf{x}_j\|^2 = 2n_k \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2;$$

multiply by $\frac{1}{2} \cdot \frac{1}{n_k}$. $T = W + B$ is the trace of the MANOVA SSP decomposition (Ch. 6). ■

Since T is fixed, minimizing W is maximizing B.

1.1 Limits of combinatorial algorithms

The number of ways to partition n items into K non-empty clusters (Stirling number of the second kind) is

$$A(n, K) = \frac{1}{K!} \sum_{k=1}^{K} (-1)^{K-k} \binom{K}{k} k^n \sim \frac{K^n}{K!}.$$

E.g. $A(19, 4) \approx 10^{10}$: exhaustive search is impossible, so greedy iterative algorithms are used.

2 K-means and K-medoids

Motivation. Minimizing within-cluster scatter exactly is combinatorially hard. K-means alternates two easy steps, assigning points and updating centres, each of which decreases the criterion, so it converges to a local optimum.

2.1 K-means

With $d(i,j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2^2$, any point $\boldsymbol{m}$ satisfies $\sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}\|^2 \geq \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, so

$$W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|_2^2 = \min_{\boldsymbol{m}_1,\ldots,\boldsymbol{m}_K} L(C \mid \boldsymbol{m}), \qquad L(C \mid \boldsymbol{m}) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \boldsymbol{m}_k\|_2^2.$$

Definition 11.3: K-means algorithm

(i) Minimize over C with $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ fixed: update $C(i) = k$ for the closest center.

(ii) Minimize over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$ with C fixed: update $\boldsymbol{m}_k$ by the new average of cluster k.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Theorem 11.4: convergence

K-means stops after finitely many iterations at a local minimum.

Proof [Supplement]. Each step minimizes L in one block of variables with the other fixed, so L is non-increasing. There are finitely many encoders C and step (ii) is unique for a given C, so while L strictly decreases no C can recur; the algorithm stops in finitely many steps. ■

The result depends on the starting values, so run several starts and keep the smallest W. K-means is scale-dependent; consider standardizing.

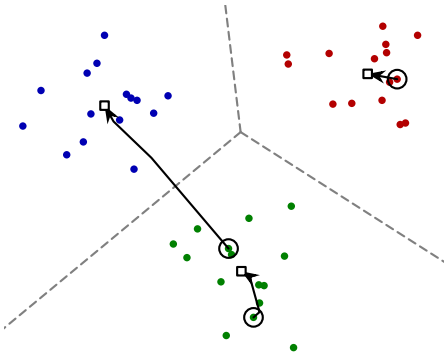


Figure. *K-means ($K = 3$) on synthetic data, 45 points from three groups. The starting centres (circles) are three observations, two of them in the bottom group. Black arrows trace the centres over the iterations until C stops changing (squares, after three updates). The dashed lines are the perpendicular bisectors of the final centres: step (i) assigns each point to the Voronoi cell of its nearest centre, so the cells are convex.*

2.2 K-medoids

Use an arbitrary dissimilarity and restrict centers to observations.

(i) Minimize over C: update $C(i) = k$ for the closest medoid $\boldsymbol{m}_k$.

(ii) Minimize over $\boldsymbol{m}_1, \ldots, \boldsymbol{m}_K$: update $\boldsymbol{m}_k$ by the item in the kth cluster that minimizes the total dissimilarity to the cluster.

(iii) Iterate (i), (ii) until $L(C \mid \boldsymbol{m})$ does not change.

Using actual observations as representatives makes it robust to outliers and usable for categorical data.

3 Hierarchical Clustering

3.1 Agglomerative

Start with singletons and repeatedly merge the two closest clusters. Between-cluster distances (linkage):

Single linkage	$d(G_1, G_2) = \min_{i\in G_1, j\in G_2} d_{ij}$
Complete linkage	$d(G_1, G_2) = \max_{i\in G_1, j\in G_2} d_{ij}$
Average linkage	$d(G_1, G_2) = \text{average}_{i\in G_1, j\in G_2} d_{ij}$

Single linkage tends to produce long chained clusters, complete linkage compact round ones. The result is drawn as a dendrogram and cut at a chosen height.

Ward's method merges, at each step, the two clusters whose union increases the within-cluster sum of squares $W(C)$ the least.

Theorem 11.5: Ward's merging cost

Let $\text{ESS}(G) = \sum_{i\in G} \|\mathbf{x}_i - \bar{\mathbf{x}}_G\|^2$. For disjoint clusters G, H of sizes n_G, n_H,

$$\text{ESS}(G \cup H) - \text{ESS}(G) - \text{ESS}(H) = \frac{n_G n_H}{n_G + n_H} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2.$$

Proof [Supplement].

Step 1. Applying $T = W + B$ (Theorem 11.2) to the set $G \cup H$ split into the two clusters G, H, with $N = n_G + n_H$ and $\bar{\mathbf{x}} = (n_G \bar{\mathbf{x}}_G + n_H \bar{\mathbf{x}}_H)/N$ the mean of $G \cup H$,

$$\text{ESS}(G \cup H) = \text{ESS}(G) + \text{ESS}(H) + n_G \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}\|^2 + n_H \|\bar{\mathbf{x}}_H - \bar{\mathbf{x}}\|^2.$$

Step 2. $\bar{\mathbf{x}}_G - \bar{\mathbf{x}} = \frac{n_H}{N}(\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H)$ and $\bar{\mathbf{x}}_H - \bar{\mathbf{x}} = -\frac{n_G}{N}(\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H)$, so the last two terms equal $\frac{n_G n_H^2 + n_H n_G^2}{N^2} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2 = \frac{n_G n_H}{N} \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|^2$. ■

So Ward's method is the greedy version of K-means' criterion: it is centroid distance weighted by cluster sizes, which penalizes merging two large clusters and favours compact clusters of similar size.

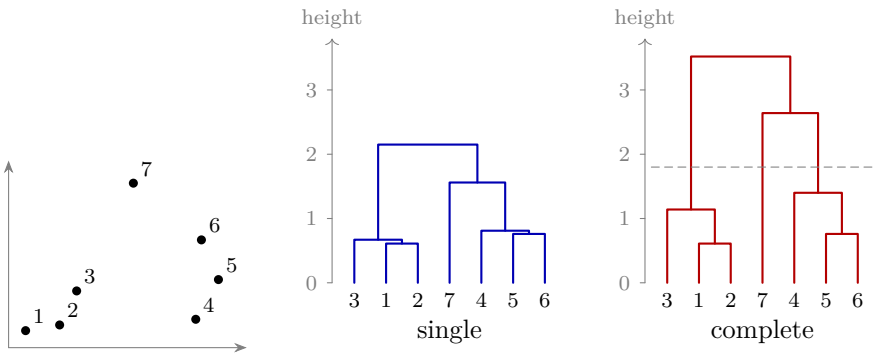


Figure. *Seven synthetic points (left) and their single- and complete-linkage dendrograms (Euclidean distance). Both merge* $\{1, 2\}$ *first at height* 0.61*, but single linkage then chains the isolated point* 7 *onto* $\{4, 5, 6\}$ *at the smallest gap* 1.56*, while complete linkage merges* 7 *only at* 2.64*. Cutting the complete-linkage tree at the dashed height gives the three clusters* $\{1, 2, 3\}$*,* $\{4, 5, 6\}$*,* $\{7\}$*. In both trees the heights increase upwards (Exercise 11.4).*

Distances between observations:

$$p\text{-norm: } d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_p = \Big(\sum_{l=1}^{d} |x_l - y_l|^p \Big)^{1/p},$$

$$\text{standardized distance: } d^2(\mathbf{x},\mathbf{y}) = \sum_{l=1}^{d} \frac{(x_l - y_l)^2}{s_l^2} = (\mathbf{x}-\mathbf{y})^\top \mathbf{D}^{-1}(\mathbf{x}-\mathbf{y}).$$

3.2 Divisive

Start from a single cluster and split repeatedly (e.g. $K = 2$ means at each step).

4 Gaussian Mixtures and the EM Algorithm

4.1 Model

$\mathbf{X}_1, \dots, \mathbf{X}_n \overset{\text{iid}}{\sim} g(\mathbf{x})$ with latent labels $\delta_1, \dots, \delta_n \overset{\text{iid}}{\sim}$, $P(\delta_i = k) = \pi_k$, and $\mathbf{X} \mid \delta = k \sim N_p(\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$:

$$\mathbf{X} \sim g(\mathbf{x}) = \sum_{k=1}^{K} \text{pdf}(\mathbf{x} \mid \delta = k) P(\delta = k) = \sum_{k=1}^{K} \pi_k g_k(\mathbf{x}),$$

$$L(\boldsymbol{\theta} \mid \mathbf{X}) = \prod_{i=1}^{n} g(\mathbf{x}_i) = \prod_{i=1}^{n} \Big\{ \sum_{k=1}^{K} g_k(\mathbf{x}_i)\pi_k \Big\},$$

giving $\hat{\pi}_k, \hat{\boldsymbol{\mu}}_k, \hat{\boldsymbol{\Sigma}}_k$. Then $\hat{P}(\delta = k \mid \mathbf{x}) = \hat{g}_k(\mathbf{x})\hat{P}(\delta = k)/\hat{g}(\mathbf{x})$ and $C : \mathcal{X} \to \{1, \dots, K\}$, $C(\mathbf{x}) = k$ iff $\hat{P}(\delta = k \mid \mathbf{x}) \geq \hat{P}(\delta = j \mid \mathbf{x})$ for all j (soft clustering). With all $\boldsymbol{\Sigma}_k = \sigma^2 \mathbf{I}$ and $\sigma \to 0$ this becomes K-means.

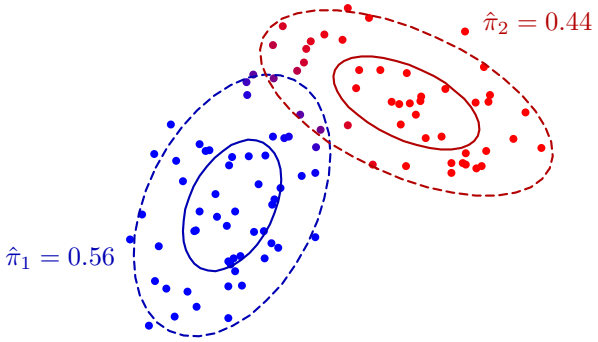


Figure. *A two-component Gaussian mixture fitted by EM to* 100 *synthetic points (*60 *and* 40 *drawn from two bivariate normals). Ellipses are the* 1- *and* 2*-Mahalanobis-distance contours of the fitted components. Each point is coloured by its responsibility* $\hat{P}(\delta = 1 \mid \mathbf{x})$*, from blue (*1*) to red (*0*); points in the overlap get intermediate colours, the soft assignment that* K*-means lacks.*

4.2 The EM algorithm ($K = 2$)

$\mathbf{X}_1, \dots, \mathbf{X}_n \sim \pi_1 \phi(\mathbf{x}; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \pi_0 \phi(\mathbf{x}; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$, $\pi_0 = 1 - \pi_1$. The observed-data log-likelihood

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log\{\pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0) + \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)\}$$

has no closed-form maximizer (log of a sum). If $\delta_1, \dots, \delta_n \in \{0, 1\}$ were observed, $L(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \prod_i \text{pdf}(\mathbf{x}_i, \delta_i \mid \boldsymbol{\theta}) = \prod_i \text{pdf}(\mathbf{x}_i \mid \delta_i, \boldsymbol{\theta}) \text{pdf}(\delta_i)$ and

$$\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n) = \sum_{i:\delta_i = 1} \log \pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) + \sum_{i:\delta_i = 0} \log \pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)$$

$$= \sum_{i=1}^{n} \left[\delta_i \log \pi_1 \phi_1(\mathbf{x}_i) + (1 - \delta_i) \log \pi_0 \phi_0(\mathbf{x}_i)\right],$$

which separates and gives closed forms such as $\hat{\pi}_1^{\mathrm{MLE}} = n_1/n$.

Definition 11.6: EM algorithm

(i) $\ell(\boldsymbol{\theta} \mid \mathbf{X}^n, \boldsymbol{\delta}^n)$ has an easy MLE, but $\delta_1, \ldots, \delta_n$ are unobserved;
(ii) iteratively approximate the maximum. At $\boldsymbol{\theta}^{(m)}$:

- **E-step**: predict δ_i by $\hat{\delta}_i = \mathrm{E}(\delta_i \mid \mathbf{x}_i, \boldsymbol{\theta}^{(m)})$, i.e. compute $Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) := \mathrm{E}[\ell(\boldsymbol{\theta} \mid D) \mid D_{\mathrm{obs}}, \boldsymbol{\theta}^{(m)}]$. Since the complete-data log-likelihood is linear in δ_i,

$$\hat{\delta}_i = \frac{\pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})}{\pi_0^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_0^{(m)}, \boldsymbol{\Sigma}_0^{(m)}) + \pi_1^{(m)} \phi(\mathbf{x}_i; \boldsymbol{\mu}_1^{(m)}, \boldsymbol{\Sigma}_1^{(m)})} \quad \text{(responsibility)}.$$

- **M-step**: $\boldsymbol{\theta}^{(m+1)} = \arg\max_{\boldsymbol{\theta}} L(\boldsymbol{\theta} \mid \mathbf{X}^n, \hat{\boldsymbol{\delta}}^n)$, i.e. $\arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$:

$$\pi_1 = \frac{\sum \hat{\delta}_i}{n}, \quad \boldsymbol{\mu}_1 = \frac{\sum \hat{\delta}_i \mathbf{x}_i}{\sum \hat{\delta}_i}, \quad \boldsymbol{\Sigma}_1 = \frac{\sum \hat{\delta}_i (\mathbf{x}_i - \boldsymbol{\mu}_1)(\mathbf{x}_i - \boldsymbol{\mu}_1)^\top}{\sum \hat{\delta}_i},$$

  and similarly for component 0 with weights $1 - \hat{\delta}_i$.

Repeat until $d(\boldsymbol{\theta}^{(m)}, \boldsymbol{\theta}^{(m+1)})$ is small.

Theorem 11.7: monotonicity of EM

$\ell(\boldsymbol{\theta}^{(m+1)}) \geq \ell(\boldsymbol{\theta}^{(m)})$.

Proof [Supplement]. $\ell(\boldsymbol{\theta}) = Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)})$ with $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) = \mathrm{E}[\log p(\boldsymbol{\delta} \mid \mathbf{X}, \boldsymbol{\theta}) \mid \mathbf{X}, \boldsymbol{\theta}^{(m)}]$. By Gibbs' (Jensen's) inequality $H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^{(m)}) \leq H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})$ (KL divergence ≥ 0). The M-step does not decrease Q, so $\ell(\boldsymbol{\theta}^{(m+1)}) - \ell(\boldsymbol{\theta}^{(m)}) = \{Q(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - Q(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} - \{H(\boldsymbol{\theta}^{(m+1)} \mid \boldsymbol{\theta}^{(m)}) - H(\boldsymbol{\theta}^{(m)} \mid \boldsymbol{\theta}^{(m)})\} \geq 0$. ■

EM may converge to a local maximum (or saddle point), and the likelihood is unbounded as some $\boldsymbol{\Sigma}_k$ becomes singular, so use several starts and regularization.

5 Choosing K and Other Topics

- **Scatter decomposition**: $W(C) = \sum_{k=1}^{K} \sum_{C(i)=k} \|\mathbf{x}_i - \bar{\mathbf{x}}_{(k)}\|^2$, $B(C) = \sum_{k=1}^{K} n_k \|\bar{\mathbf{x}}_{(k)} - \bar{\mathbf{x}}\|^2$. W always decreases as K grows, so it cannot choose K by itself (elbow method: where the decrease flattens).
- Indices: $\mathrm{CI}(K, c) = W(K, c)/T$ (smaller is better) and Calinski–Harabasz $\mathrm{CH}(K, c) = \dfrac{B(K, c)/(K-1)}{W(K, c)/(n-K)}$ (larger is better; the analogue of the one-way MANOVA F).
- **Gap statistic**: compare with the expected $\log W_K^*$ under a reference distribution (uniform over the data range),

$$\mathrm{Gap}(K) = \mathrm{E}^*[\log W_K^*] - \log W_K, \qquad s(K) = \text{(adjusted) s.d. of } \log W_K^*,$$

  and choose the smallest K with $\mathrm{Gap}(K) \geq \mathrm{Gap}(K+1) - s(K+1)$.
- **Clustering variables**: define $d(v_i, v_j) = 1 - r_{ij}$ or $1 - r_{ij}^2$ and apply hierarchical clustering.

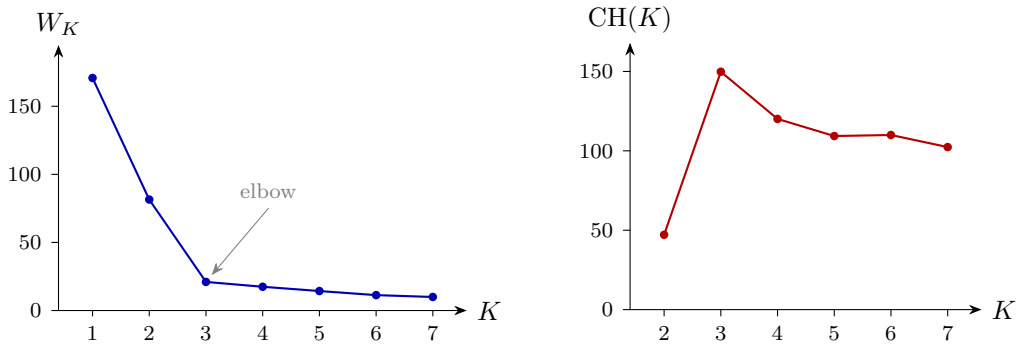


Figure. *Choosing K for the K-means data of the earlier figure (best of 20 random starts for each K). Left: W_K keeps decreasing, but the decrease flattens after $K = 3$ (elbow). Right: the Calinski–Harabasz index, which divides the between and within scatter by their degrees of freedom, has its maximum at $K = 3$.*

6 Selected exercises

Motivation. The exercises prove facts used above without proof: the count of partitions, the closed-form M-step, the unbounded mixture likelihood, and the monotone merge heights that make a dendrogram well defined.

Exercise 11.1 (Counting partitions)

Show that the number of partitions of $n \geq 1$ items into K non-empty clusters is

$$A(n, K) = \frac{1}{K!} \sum_{k=0}^{K} (-1)^{K-k} \binom{K}{k} k^n,$$

and that $A(n, K)\, K!/K^n \to 1$ as $n \to \infty$ for fixed K.

Solution [Supplement].

Step 1 (encoders). A partition into K non-empty blocks together with a labelling of the blocks by $1, \dots, K$ is the same as a surjective encoder $C : \{1, \dots, n\} \to \{1, \dots, K\}$. Each partition has $K!$ labellings, so $A(n, K) = N/K!$ with N the number of surjections.

Step 2 (inclusion–exclusion). Let E_j be the set of encoders that never use label j. For $J \subseteq \{1, \dots, K\}$ with $|J| = m$, $\left|\bigcap_{j \in J} E_j\right| = (K - m)^n$. By inclusion–exclusion,

$$N = \sum_{J} (-1)^{|J|} \Big| \bigcap_{j \in J} E_j \Big| = \sum_{m=0}^{K} (-1)^m \binom{K}{m} (K - m)^n,$$

and the substitution $k = K - m$ gives the formula. The $k = 0$ term vanishes for $n \geq 1$, which gives the form used above.

Step 3 (asymptotics). $K!\, A(n, K)/K^n = \sum_{k=0}^{K} (-1)^{K-k} \binom{K}{k} (k/K)^n \to 1$, since $(k/K)^n \to 0$ for $k < K$. For example $A(19, 4) \approx 4^{19}/24 \approx 1.1 \times 10^{10}$. ■

Exercise 11.2 (Deriving the M-step)

Given responsibilities $\hat{\delta}_i \in (0, 1)$, let

$$Q(\boldsymbol{\theta}) = \sum_{i=1}^{n} \left[\hat{\delta}_i \log\{\pi_1 \phi(\mathbf{x}_i; \boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)\} + (1 - \hat{\delta}_i) \log\{\pi_0 \phi(\mathbf{x}_i; \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0)\} \right], \qquad \pi_0 = 1 - \pi_1.$$

Assuming the weighted scatter matrices below are positive definite, show that Q is maximized by the M-step formulas of the EM algorithm.

Solution [Supplement].

Step 1 (separation). With $N_1 = \sum_i \hat\delta_i \in (0, n)$ and $N_0 = n - N_1$,

$$Q = \{N_1 \log \pi_1 + N_0 \log(1-\pi_1)\} + Q_1(\boldsymbol\mu_1, \boldsymbol\Sigma_1) + Q_0(\boldsymbol\mu_0, \boldsymbol\Sigma_0), \qquad Q_1 = \sum_i \hat\delta_i \log \phi(\mathbf{x}_i; \boldsymbol\mu_1, \boldsymbol\Sigma_1),$$

and Q_0 likewise with weights $1 - \hat\delta_i$; the three parts have separate parameters.

Step 2 (π_1). $N_1 \log \pi + N_0 \log(1-\pi)$ is strictly concave on $(0,1)$ with derivative $N_1/\pi - N_0/(1-\pi)$, which vanishes only at $\pi = N_1/n$.

Step 3 ($\boldsymbol\mu_1$). $-2Q_1 = N_1 p \log 2\pi + N_1 \log|\boldsymbol\Sigma_1| + \sum_i \hat\delta_i (\mathbf{x}_i - \boldsymbol\mu_1)^\top \boldsymbol\Sigma_1^{-1} (\mathbf{x}_i - \boldsymbol\mu_1)$. With $\boldsymbol m_1 = \sum_i \hat\delta_i \mathbf{x}_i / N_1$ we have $\sum_i \hat\delta_i (\mathbf{x}_i - \boldsymbol m_1) = \mathbf{0}$, so

$$\sum_i \hat\delta_i (\mathbf{x}_i - \boldsymbol\mu_1)^\top \boldsymbol\Sigma_1^{-1} (\mathbf{x}_i - \boldsymbol\mu_1) = \sum_i \hat\delta_i (\mathbf{x}_i - \boldsymbol m_1)^\top \boldsymbol\Sigma_1^{-1} (\mathbf{x}_i - \boldsymbol m_1) + N_1 (\boldsymbol m_1 - \boldsymbol\mu_1)^\top \boldsymbol\Sigma_1^{-1} (\boldsymbol m_1 - \boldsymbol\mu_1),$$

minimized by $\boldsymbol\mu_1 = \boldsymbol m_1$ whatever $\boldsymbol\Sigma_1$ is (profiling, Theorem 2.26).

Step 4 ($\boldsymbol\Sigma_1$). With $\mathbf{V}_1 = \sum_i \hat\delta_i (\mathbf{x}_i - \boldsymbol m_1)(\mathbf{x}_i - \boldsymbol m_1)^\top / N_1 \succ 0$, the profiled criterion is $N_1\{\log|\boldsymbol\Sigma_1| + \operatorname{tr}(\boldsymbol\Sigma_1^{-1}\mathbf{V}_1)\} + \text{const}$. Let $\mathbf{B} = \mathbf{V}_1^{1/2} \boldsymbol\Sigma_1^{-1} \mathbf{V}_1^{1/2} \succ 0$ with eigenvalues β_j. Then

$$\log|\boldsymbol\Sigma_1| + \operatorname{tr}(\boldsymbol\Sigma_1^{-1}\mathbf{V}_1) - \log|\mathbf{V}_1| - p = \operatorname{tr}\mathbf{B} - \log|\mathbf{B}| - p = \sum_{j=1}^{p} (\beta_j - \log\beta_j - 1) \geq 0$$

by Theorem 2.15, with equality iff all $\beta_j = 1$, i.e. $\mathbf{B} = \mathbf{I}$, i.e. $\boldsymbol\Sigma_1 = \mathbf{V}_1$.

Step 5. The same argument with weights $1 - \hat\delta_i$ gives $\boldsymbol\mu_0$ and $\boldsymbol\Sigma_0$. These are the formulas of the EM algorithm: weighted versions of the MLEs of Theorem 4.7. ■

Exercise 11.3 (The mixture likelihood is unbounded)

Let $x_1, \ldots, x_n$ be distinct reals and $\ell(\boldsymbol\theta) = \sum_i \log\{\pi\phi(x_i; \mu_1, \sigma_1^2) + (1-\pi)\phi(x_i; \mu_0, \sigma_0^2)\}$.
(a) Show $\sup_{\boldsymbol\theta} \ell = +\infty$.
(b) Show that if the variances are constrained to be equal, $\sigma_0 = \sigma_1 = \sigma$, and $n \geq 3$, then ℓ is bounded above.

Solution [Supplement].

(a) Fix $\pi \in (0,1)$, μ_0 and σ_0, and set $\mu_1 = x_1$. The $i = 1$ term is at least $\log\{\pi/(\sqrt{2\pi}\sigma_1)\} \to \infty$ as $\sigma_1 \downarrow 0$, while each term $i \geq 2$ is at least $\log\{(1-\pi)\phi(x_i; \mu_0, \sigma_0^2)\}$, a finite constant. Hence $\ell \to \infty$: a component collapses onto one observation, and no global MLE exists.

(b)

Step 1. Let $\delta = \min_{i \neq j} |x_i - x_j| > 0$. A mean μ_k is within distance $< \delta/2$ of at most one x_i, so for any (μ_0, μ_1) at least $n - 2 \geq 1$ observations, say x_{i_0}, satisfy $|x_{i_0} - \mu_k| \geq \delta/2$ for $k = 0, 1$.

Step 2. Every term is at most $\log\{1/(\sqrt{2\pi}\sigma)\}$, and the i_0 term is at most $\log\{1/(\sqrt{2\pi}\sigma)\} - \delta^2/(8\sigma^2)$. So

$$\ell \leq u(\sigma) := -n \log(\sqrt{2\pi}\,\sigma) - \frac{\delta^2}{8\sigma^2}.$$

Step 3. u is continuous on $(0, \infty)$ and tends to $-\infty$ as $\sigma \downarrow 0$ and as $\sigma \to \infty$, so $\sup_\sigma u(\sigma) < \infty$, which bounds ℓ. This is why EM is run with constrained or regularized covariances. ■

Exercise 11.4 (Monotone merge heights)

In agglomerative clustering let h_t be the linkage distance of the two clusters merged at step t.
(a) Show $h_1 \le h_2 \le \cdots$ for single, complete and average linkage.
(b) Show by example that centroid linkage, $d(G,H) = \|\bar{\mathbf{x}}_G - \bar{\mathbf{x}}_H\|$, can produce $h_2 < h_1$.

Solution [Supplement].

(a)

Step 1 (update rule). If G_1, G_2 are merged into $G = G_1 \cup G_2$, then for every other cluster H

$$d(G,H) = \min\{d(G_1,H), d(G_2,H)\}, \quad \max\{d(G_1,H), d(G_2,H)\}, \quad \frac{|G_1|d(G_1,H) + |G_2|d(G_2,H)}{|G_1| + |G_2|}$$

for single, complete and average linkage respectively, since the pairs $(i,j) \in G \times H$ are the union of $G_1 \times H$ and $G_2 \times H$.

Step 2. At step t the merged pair has the smallest distance among current clusters, so $d(G_1,H), d(G_2,H) \ge h_t$ and all distances between untouched clusters are $\ge h_t$. A minimum, maximum or weighted average of numbers $\ge h_t$ is $\ge h_t$, so every distance after the merge is $\ge h_t$, and $h_{t+1} \ge h_t$.

(b) Take the vertices of an equilateral triangle with side 1. First two vertices merge at $h_1 = 1$; their centroid is the midpoint of a side, at distance $\sqrt{3}/2$ from the third vertex, so $h_2 = \sqrt{3}/2 < h_1$ and the dendrogram has an inversion. ■